

Contents

2	Field of Complex Numbers	2
2.1	Definitions and general notations	3
2.2	The complex plane	3
2.3	Conjugate of a complex number	4
2.4	Modulus of a complex number	5
2.5	Argument	5
2.6	Trigonometric form	6
2.7	Moivre's formula	6
2.8	n-th root of a Complex Number	8
2.9	Square root of a Complex Number	9
2.10	Solving second degree equations in $\mathbb{C}$	10

Chapter

2

Field of Complex Numbers

Contents

2.1	Definitions and general notations	3
2.2	The complex plane	3
2.3	Conjugate of a complex number	4
2.4	Modulus of a complex number	5
2.5	Argument	5
2.6	Trigonometric form	6
2.7	Moivre's formula	6
2.8	n-th root of a Complex Number	8
2.9	Square root of a Complex Number	9
2.10	Solving second degree equations in $\mathbb{C}$	10

We know that the square of a real number is always non-negative e.g. $(4)^2 = 16$ and $(-4)^2 = 16$. Therefore, the square root of 16 is (± 4) . What about the square root of a negative number? It is clear that a negative number can not have a real square root. So we need to extend the system of real numbers to a system in which we can find out the square roots of negative numbers. Euler (1707 - 1783) was the first mathematician to introduce the symbol i (iota) for the positive square root of -1 i.e., $i = \sqrt{-1}$.

2.1 Definitions and general notations

Definition 2.1.1. *The set of complex numbers $\mathbb{C}$ is written in the form:*

$$\mathbb{C} = \{z = x + iy, \quad x, y \in \mathbb{R}, \quad i^2 = -1\}$$

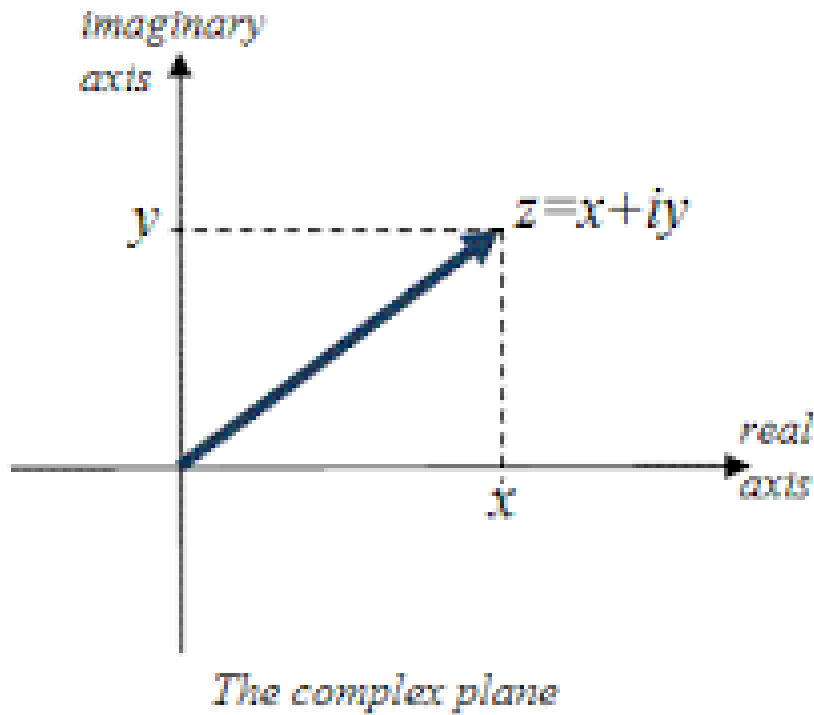
where

- $x = \text{Re}(z)$ is the real part of z .
- $y = \text{Im}(z)$ is its imaginary part.
- If $y = 0$, z is said to be real.
- If $x = 0$, z is said to be imaginary.

2.2 The complex plane

Just as real numbers can be visualized as points on a line, complex numbers can be visualized as points in a plane: plot $z = x + iy$ at the point (x, y) .

Example 2.2.1. $1 + 3i, \quad \sqrt{2} - \pi i, \quad i, \quad 4, \quad \sqrt{6}i, \quad \frac{2}{3}i, \quad 1 - \frac{4}{3}i$



2.3 Conjugate of a complex number

Definition 2.3.1. Let $z = x + iy$ be a complex number, the number $\bar{z} = x - iy$ is called the conjugate of z .

Example 2.3.1. , if $z = 3 + 2i$ then $\bar{z} = 3 - 2i$,

if $z = -4 - i$ then $\bar{z} = -4 + i$,

if $z = 5 - 3i$ then $\bar{z} = 5 + 3i$.

Properties 2.3.1. 1. $\overline{(z + z')} = \bar{z} + \bar{z}'$

2. $\overline{(z \cdot z')} = \bar{z} \cdot \bar{z}'$.

3. $z + \bar{z} = 2x = 2\operatorname{Re}(z)$.

4. $z - \bar{z} = 2i\operatorname{Im}(z)$.

$$5. \bar{\bar{z}} = z.$$

2.4 Modulus of a complex number

Definition 2.4.1. We call modulus of a complex number $z = x + iy$, the positive real number $|z|$, defined by:

$$|z| = \sqrt{z \cdot \bar{z}} = \sqrt{x^2 + y^2}$$

Properties 2.4.1. For all $z, z' \in \mathbb{C}$, we have

1. $|z| \geq 0$.
2. $|z| = 0 \Leftrightarrow z = 0$,
3. $|z \cdot z'| = |z| \cdot |z'|$,
4. $|z + z'| \leq |z| + |z'|$, (Triangle inequality).
5. $|\frac{z}{z'}| = \frac{|z|}{|z'|}$, $z' \neq 0$.
6. $|\frac{1}{z}| = \frac{1}{|z|}$, $z \neq 0$.

2.5 Argument

Definition 2.5.1. Let $z = x + iy$ be a non zero complex number. An argument of z denoted by $\arg(z)$ is any real number θ such that

$$\arg(z) = \begin{cases} \cos \theta = \frac{x}{|z|}, \\ \sin \theta = \frac{y}{|z|}. \end{cases}$$

Properties 2.5.1. Let z_1 and z_2 be two complex numbers

$$1. \arg(z_1 z_2) = \arg(z_1) + \arg(z_2).$$

$$2. \arg(z^n) = n \arg(z).$$

$$3. \arg\left(\frac{1}{z}\right) = -\arg(z).$$

$$4. \arg(\bar{z}) = -\arg(z).$$

$$5. \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2).$$

2.6 Trigonometric form

Let $z = x + iy$, $|z| = \sqrt{x^2 + y^2}$, and $\theta = \arg(z)$.

We have

$$\begin{cases} x = |z| \cos \theta, \\ y = |z| \sin \theta. \end{cases}$$

The Trigonometric form of z is given by

$$z = |z| (\cos \theta + i \sin \theta) = |z| e^{i\theta}$$

. This representation is very useful for the multiplication and division of complex numbers:

- $z_1 \times z_2 = |z_1| e^{i\theta_1} \times |z_2| e^{i\theta_2} = |z_1| \cdot |z_2| e^{i(\theta_1 + \theta_2)}.$
- $\frac{z_1}{z_2} = \frac{|z_1| e^{i\theta_1}}{|z_2| e^{i\theta_2}} = \frac{|z_1|}{|z_2|} e^{i(\theta_1 - \theta_2)}.$

2.7 Moivre's formula

For positive integers n we have the **Moivre's formula**:

$$(e^{i\theta})^n = e^{in\theta} \Leftrightarrow (\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

where

$$\begin{cases} \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \\ \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \end{cases}$$

Application:

By developing the Moivre formula using the Newton binomial formula:

$$(\cos \theta + i \sin \theta)^n = \sum_{k=0}^n C_n^k (\cos \theta)^{n-k} (i \sin \theta)^k.$$

Where $C_n^k = \frac{n!}{k!(n-k)!}$, $C_n^n = \frac{n!}{n!(0)!} = 1$, and $C_n^0 = \frac{n!}{0!(n)!} = 1$.

We have

$$\begin{aligned} (\cos \theta + i \sin \theta)^n &= C_n^0 (\cos \theta)^n (i \sin \theta)^0 + C_n^1 (\cos \theta)^{n-1} (i \sin \theta)^1 + \dots + C_n^k (\cos \theta)^{n-k} (i \sin \theta)^k + \\ &\dots + C_n^n (\cos \theta)^0 (i \sin \theta)^n. \end{aligned}$$

So, we get

The real part:

$$(\cos n\theta) = (\cos \theta)^n - C_n^2 (\cos \theta)^{n-2} (\sin \theta)^2 + C_n^4 (\cos \theta)^{n-4} (\sin \theta)^4 + \dots$$

and

The imaginary part:

$$(\sin n\theta) = C_n^1 (\cos \theta)^{n-1} (\sin \theta)^1 - C_n^3 (\cos \theta)^{n-3} (\sin \theta)^3 + \dots$$

Example 2.7.1. For $n = 3$:

$$\begin{aligned} (\cos(\theta) + i \sin(\theta))^3 &= \sum_{k=0}^3 C_3^k (\cos(\theta))^{3-k} (i \sin(\theta))^k \\ &= \cos^3(\theta) + 3i \cos^2(\theta) \sin(\theta) - 3 \cos(\theta) \sin^2(\theta) - i \sin^3(\theta). \end{aligned}$$

By identifying the real and imaginary parts, we deduce that:

$$\cos(3\theta) = \cos^3(\theta) - 3 \cos(\theta) \sin^2(\theta), \quad \text{and} \quad \sin(3\theta) = 3 \cos^2(\theta) \sin(\theta) - \sin^3(\theta).$$

2.8 n-th root of a Complex Number

Definition 2.8.1. A complex number z is an n – th root of w if:

$$z^n = w$$

Suppose that $z = r(\cos(\theta) + i \sin(\theta))$ is an n – th root of $w = \rho(\cos(\alpha) + i \sin(\alpha))$.

By using the Moivre's formula, we obtain

$$z^n = w \implies r^n (\cos(n\theta) + i \sin(n\theta)) = \rho (\cos(\alpha) + i \sin(\alpha))$$

So

$$\begin{cases} r^n = \rho \\ n\theta = \alpha + 2k\pi, \quad 0 \leq k \leq n-1. \end{cases}$$

Then

$$\begin{cases} r = \sqrt[n]{\rho} \\ \theta = \frac{\alpha + 2k\pi}{n}, \quad 0 \leq k \leq n-1 \end{cases}$$

thus, the n – th roots of the complex number were

$$z = \sqrt[n]{\rho} \left(\cos\left(\frac{\alpha + 2k\pi}{n}\right) + i \sin\left(\frac{\alpha + 2k\pi}{n}\right) \right), \quad 0 \leq k \leq n-1$$

Example 2.8.1. Find the cube roots of the complex number $w = 1$ ie ($z^3 = 1$)

we have: $w = 1 = \cos(2k\pi) + i \sin(2k\pi)$ and let $z \in \mathbb{C}$, $z = r(\cos \theta + i \sin \theta)$

$$z^3 = 1 \implies r^3 (\cos(3\theta) + i \sin(3\theta)) = \cos(2k\pi) + i \sin(2k\pi)$$

So

$$\begin{cases} r^3 = 1 \\ 3\theta = 2k\pi \end{cases} \implies \begin{cases} r = 1 \\ \theta = \frac{2k\pi}{3}, \end{cases} \quad 0 \leq k \leq 2$$

Then

$$z = \cos\left(\frac{2k\pi}{3}\right) + i \sin\left(\frac{2k\pi}{3}\right), \quad 0 \leq k \leq 2$$

So

$$\begin{cases} z_1 = \cos 0 + i \sin 0 = 1 \\ z_2 = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \\ z_3 = \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i. \end{cases}$$

2.9 Square root of a Complex Number

Example 2.9.1. Determine the square roots of the complex number $w = 3 + 4i$

❶ $|w| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$

❷ Let $z \in \mathbb{C}$ such that $z = x + iy$ and $z^2 = w$

$$z^2 = w \iff (x + iy)^2 = 3 + 4i \iff x^2 - y^2 + 2ixy = 3 + 4i$$

So

$$\begin{cases} x^2 - y^2 = 3 \\ 2xy = 4 \end{cases}$$

and we have

$$|w| = |z^2| = |z|^2 = x^2 + y^2 = 5.$$

So $x = +2, -2$ and $y = +1, -1$

Then, the square roots of w are

$$\begin{cases} z_1 = 2 + i \\ z_2 = -2 - i \end{cases}$$

2.10 Solving second degree equations in $\mathbb{C}$

Given the equation (E):

$$az^2 + bz + c = 0, \text{ where } a, b, c \in \mathbb{R} \text{ } a \neq 0 \text{ and } z \in \mathbb{C}$$

To solve it, we compute the discriminant

$$\Delta = b^2 - 4ac$$

and we distinguish three cases:

- ❶ 1 st case: if $\Delta > 0$ then, the equation (E) has two distinct real solutions:

$$z_1 = \frac{-b - \sqrt{\Delta}}{2a}, \quad z_2 = \frac{-b + \sqrt{\Delta}}{2a}$$

- ❷ 2 nd case: if $\Delta = 0$ then, the equation (E) has one double root:

$$z = \frac{-b}{2a}$$

③ 3 rd case: if $\Delta < 0$ then, the equation (E) has two distinct complex solutions:

$$z_1 = \frac{-b - i\sqrt{-\Delta}}{2a}, \quad z_2 = \frac{-b + i\sqrt{-\Delta}}{2a}$$

Remark 2.10.1. *If the coefficients a, b, c are complex numbers, then Δ may also be a complex number and in that case it is necessary to compute its complex square roots.*

Example 2.10.1. *Solve in $\mathbb{C}$ the equation (E):*

$$z^2 - (7 + i)z + 12 + 3i = 0$$